TRACE — Temporal-Relational Agentic Correlation Engine

Product, Technical & Software Requirements · Oct 16, 2026 · @Sk

0. About these documents

These three documents define TRACE v1: a pilot-ready build for one Kerala Cyber Police Station, delivered in 9 months. Seven decisions below must be settled before Month 1 sprint planning; the first one blocks architecture.

Document	Answers	Main readers
Part 1 — PRD (Product Requirements)	What we are building, for whom, and why; what success looks like	Founders, Cyberdome partners, mentors, evaluators
Part 2 — TRD (Technical Requirements)	How it is built: architecture, agents, data, APIs, security, deployment	Engineers
Part 3 — SRS (Software Requirements Specification)	Numbered, testable requirements with acceptance criteria and traceability	Engineers, QA, security auditors

Conventions. "Shall" means mandatory. Priority uses MoSCoW: **M** (Must, v1 pilot), **S** (Should, v1 if time allows), **C** (Could, post-pilot). Requirement IDs (FR-INT-01, NFR-SEC-03) are stable; if a requirement is dropped, mark it withdrawn and don't reuse its ID.

Decisions to settle before development

#	Decision	Why it matters	Recommendation
D1	**Where the LLM runs.** The pitch says "no evidence leaves the agency's environment", but the stack calls the Anthropic Claude API, which is a cloud service.	This contradiction will be the first question a police IT or security reviewer asks. It also shapes the architecture and the cost model.	Put every model call behind one internal **LLM Gateway** (TRD §2.4). For the pilot, use Claude through an India-region cloud deployment under a written data-processing agreement with the agency, and send only text extracts, never raw media. Keep an on-premise open-weight model as the fallback for air-gapped stations. Reword the pitch to "evidence files never leave the agency; only redacted text is sent for processing" until this is settled.
D2	**Claude model version.** The form names `claude-sonnet-4-6`.	Model versions get retired. Hard-coding one creates a migration risk partway through the pilot.	Default to the current model, `claude-sonnet-5`, set as a configuration value. Pin one version per release and re-run the evaluation set (TRD §2.10) before any upgrade.
D3	**Legal citation for e-FIR and Zero FIR**	The pitch cites BNSS. A wrong section number undermines credibility.	Believed to be **BNSS 2023, Section 173** (an e-FIR must be signed within 3 days). Have a legal mentor verify this before submission (Appendix B).
D4	**Who operates TRACE at the counter**	Officer-only use is simpler. Letting complainants upload files themselves through a link or kiosk adds identity, consent and abuse controls.	v1 is **officer-operated**: the officer uploads with the complainant present. Complainant self-upload (by QR link) is a post-pilot item.

#	Decision	Why it matters	Recommendation
D5	**Integration with CCTNS or the Kerala e-FIR system**	Direct integration needs government approval and API access that won't be ready in 9 months.	v1 exports a draft FIR as PDF and JSON for the officer to transcribe or attach. API integration is a phase-2 item.
D6	**Source of the labelled manipulation dataset**	Without one, false-positive rates can't be stated honestly.	Combine public image-forensics datasets with a synthetic set of WhatsApp-recompressed screenshots made in-house. Record exactly where each part came from.
D7	**Malayalam speech-recognition model**	General-purpose Whisper models are known to be weak on Malayalam.	Benchmark Whisper large-v3 against Indic fine-tuned speech models on 30+ real complaint recordings in Month 1, then choose by word error rate (WER).

Part 1 — Product Requirements Document (PRD)

1.1 Summary

TRACE is an AI case-assembly assistant for the police station counter. It turns a pile of raw complaint evidence and a spoken Malayalam or English account into a structured draft FIR. The draft contains an ordered narrative, an evidence manifest with authenticity flags, advisory BNS section suggestions with citations, and a list of questions to ask **while the complainant is still present**. The officer reviews, edits and files. TRACE never files anything itself.

The beachhead is cyber-fraud and financial-crime complaints at Kerala Cyber Police Stations. These complaints carry the most evidence, depend most on getting the order of events right, and are increasingly mixed with synthetic media.

1.2 Problem

The FIR is the foundation of an Indian criminal case, yet it is still put together by hand under time pressure. Four failures keep recurring:

Failure	What happens today	Consequence
Incomplete first reports	Missing details go unnoticed until the investigating officer opens the file weeks later	Screenshots have been deleted and memories have faded by the time anyone asks
No chronology	Evidence arrives as an unordered pile	In fraud cases the order of transactions *is* the offence, and it is rarely reconstructed properly
Unverified media	There is no way to separate an authentic screenshot from a synthetic one, and forensic labs are months behind	Almost nothing is referred, and fabricated evidence goes unexamined
Language loss	The complainant speaks Malayalam and the FIR is written in English through an unrecorded translation	Names and specifics are lost, and the translation can't be audited

Today this takes 45–90 minutes per complaint, by the team's own estimate. A baseline study in pilot Month 1 will confirm the figure.

1.3 Goals and non-goals

Goals for v1 (pilot)

1. Cut the officer's time to assemble an FIR for an evidence-heavy cyber complaint by at least 50% against the measured baseline.
2. Surface missing information at intake, so more gaps are closed before the complainant leaves.
3. Give the forensic lab a ranked referral list with stated confidence, in place of today's all-or-nothing referrals.
4. Start the chain of custody at the counter, with tamper-evident hashing of every file and every action.
5. Keep the original Malayalam visible and auditable next to the English.

Non-goals

- Deciding whether media is authentic. TRACE flags files for review; it never gives a verdict.
- Scoring the complainant's credibility or detecting deception, in any form.
- Filing or submitting an FIR automatically.
- Investigation after the FIR is registered, case management, or court filings.

- Replacing CCTNS or the state e-FIR portal.

1.4 Users and personas

Persona	Role	Needs from TRACE	Main worry
FIR writer (primary)	Constable or head constable who drafts the FIR at the counter	Gets from a pile of evidence to a clean English draft fast, with a checklist of what to ask	"Will this slow me down or make me look wrong?"
Station House Officer (SHO) (primary)	Approves and signs the FIR, accountable for its quality	Can trust the draft, see the evidence flags, and act on advisory sections with citations	Legal accountability; AI making claims it can't back up
Investigating Officer (IO) (secondary)	Picks up the case after registration	A complete file, a timeline, the hashed evidence and an open-questions log	Gaps found too late
Complainant (indirect)	Often a woman, a senior citizen or a fraud victim, under stress	Is heard in their own language, is asked the right questions once, and leaves with a reference	Being disbelieved; repeating a traumatic account
Forensic analyst (secondary)	State Forensic Science Laboratory or Cyberdome analyst	A short, ranked list of files to examine, with the reasons and the original hashes	Noise; referrals without a clear reason
Station administrator / auditor	Manages users, reviews audit logs	User management, a complete audit trail, export for inquiries	Data leaks; actions that can't be traced

1.5 Core journey at the counter

1. The officer opens a new case and records the complainant's details and consent.
2. The officer uploads the evidence bundle: screenshots, chat exports, bank SMS or statements, call logs, photos and audio. Each file is hashed into the custody chain the moment it arrives.

3. The complainant narrates the incident by voice in Malayalam or English. TRACE transcribes it and shows the original next to the English translation.
4. TRACE runs triage and metadata extraction on every file, then builds the timeline and runs the gap checks. Progress is shown live.
5. The officer works through the **gap questions** with the complainant, adding answers or more files. Affected steps re-run.
6. The officer reviews the draft FIR: edits the narrative, accepts or rejects suggested BNS sections, and confirms the forensic referral list.
7. The SHO approves the draft. TRACE exports a PDF with a QR verification code, a JSON file and the evidence manifest. The officer files the FIR through the official channel.

1.6 User stories

ID	As a…	I want to…	So that…	Priority
US-01	FIR writer	upload 40+ files in one go, including duplicates	I don't have to sort them first	M
US-02	FIR writer	see duplicate screenshots collapsed into one	the manifest stays readable	M
US-03	FIR writer	record the complainant's account in Malayalam and see an English translation beside the original	nothing is lost in translation	M
US-04	FIR writer	see all events ordered on one timeline with the source of each	I can write the narrative in the right order	M
US-05	FIR writer	get a list of missing information phrased as neutral questions	I can ask them before the complainant leaves	M
US-06	FIR writer	edit any part of the draft narrative	the final wording is mine	M
US-07	SHO	see suggested BNS sections with the matching facts and the statute text	I can decide quickly and defensibly	M
US-08	SHO	see which files are flagged for forensic review, with confidence and limitations	I refer only what matters	M

ID	As a…	I want to…	So that…	Priority
US-09	SHO	approve a draft and export a QR-verifiable PDF	the record can be checked for tampering later	M
US-10	IO	open a case and see its timeline, evidence hashes and the history of resolved and open gaps	I start with a complete file	S
US-11	Forensic analyst	receive a ranked referral list with original hashes	I examine the right three files out of ninety	S
US-12	Auditor	view the full audit trail for a case and verify the hash chain	I can prove nothing was altered	M
US-13	Administrator	create users with roles and disable them	access follows staff changes	M
US-14	Complainant	receive a printed or SMS acknowledgement with a case reference	I know my complaint was recorded	C
US-15	Legal aid worker	use a light build to assemble a pre-litigation case bundle	the same tools help pre-filing	C

1.7 Feature scope

ID	Feature	v1 scope	Priority
F1	Case intake and evidence upload	Multi-file upload, file-type allowlist, 50 MB per file, SHA-256 on arrival, consent record	M
F2	Voice narration and translation	Malayalam and English speech-to-text, translation to English, original kept side by side, officer correction	M
F3	Content Triage Agent	Classify evidence type, perceptual-hash deduplication, frequency-domain manipulation screening, ranked referral list	M
F4	Metadata Agent	EXIF, GPS and timestamp extraction, offline geocoding, conflict flags	M

ID	Feature	v1 scope	Priority
F5	Timeline Agent	Pull entities, dates, amounts and events from all sources and merge them into one ordered timeline	M
F6	Gap Hunter Agent	Completeness and consistency checks on the case graph, neutral questions for the officer	M
F7	Draft FIR builder	Narrative, evidence manifest, advisory BNS suggestions with citations, officer edits	M
F8	Chain of custody and audit	Hash chain over evidence and every action, chain verification	M
F9	Export	PDF with QR verification code, JSON, evidence manifest, forensic referral sheet	M
F10	Users and roles	Login, roles (writer, SHO, IO, analyst, admin), session timeout	M
F11	Timeline view	Interactive chronology built with vis-timeline	M
F12	Electronic-evidence certificate helper	Pre-filled hash details for a BSA Section 63 certificate (to be verified, Appendix B)	S
F13	IO hand-off view	Read-only case file for the investigating officer	S
F14	Complainant self-upload by QR link	—	C
F15	CCTNS / e-FIR integration	—	C
F16	Other languages (Tamil, Kannada, Hindi)	—	C

1.8 Product principles

1. **The officer files, not the AI.** Every suggestion is advisory, cited and editable. Nothing submits automatically.

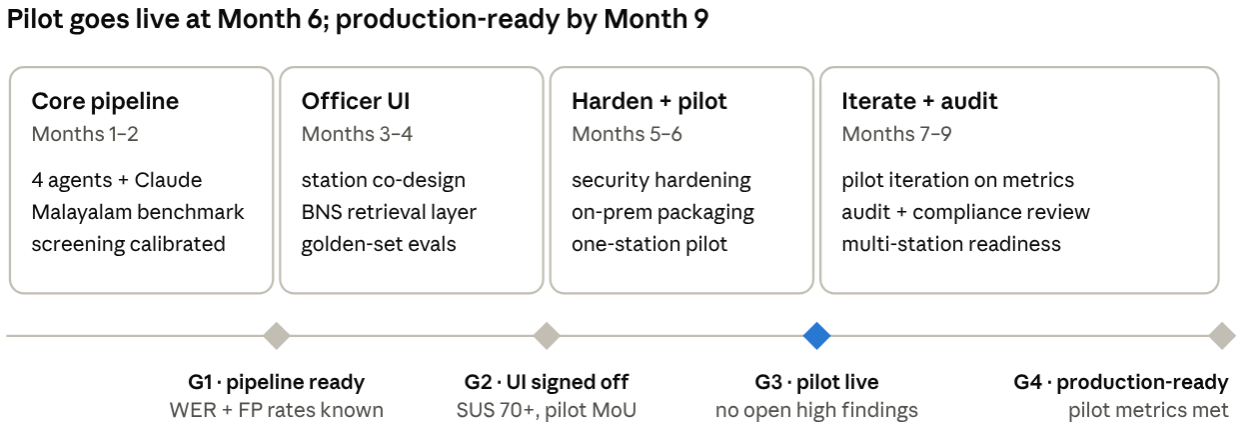
2. **Triage flags; it does not judge.** Manipulation screening gives a review recommendation with confidence and stated limits. Frequency analysis catches crude tampering and misses sophisticated forgery, and the product says so in the interface.
3. **Gap Hunter never scores credibility.** It reports what information is *absent*, in neutral language, to the officer. Inconsistent recall is a known trauma response, not a sign of deception.
4. **The original language is never silently replaced.** Translation is a visible, auditable layer.
5. **Missing metadata isn't a red flag.** WhatsApp and most messaging apps strip EXIF data. Absent metadata is shown as "not available", never as suspicious.
6. **Degrade gracefully.** If a model call fails or times out, the officer can still finish the FIR by hand, with whatever results are complete.

1.9 Success metrics

Metric	Baseline	Pilot target (proposed)	How measured
Officer minutes per FIR (evidence-heavy cyber complaint)	45–90 min (team estimate; measure in Month 1)	50% or more below the measured baseline	Timestamps from case open to SHO approval, against a timed manual control group
Gaps closed at intake	Not tracked today	70% or more of Gap Hunter questions answered before the complainant leaves	Gap status at approval
Gaps found later by the IO	Not tracked today	50% fewer than the control group	IO survey at 30 days
Forensic referral precision	All or nothing today	60% or more of referred files confirmed worth examining by the lab	Lab feedback on the referral list
Malayalam transcription quality	—	WER of 25% or less on the benchmark set; officer-corrected terms logged	Benchmark set plus correction logs

Metric	Baseline	Pilot target (proposed)	How measured
Share of the draft narrative kept by officers	—	70% or more of the narrative kept with only light edits	Edit-distance between draft and final
Officer satisfaction	—	SUS (System Usability Scale) score of 70 or more	Survey at Month 2 and Month 4 of the pilot

1.10 Release plan



9-month roadmap · 4 phases, 4 gates

Each gate must pass before the next phase starts. Decision D1 (where the LLM runs) has to be closed before G1. On the lean ₹6L budget, the external security audit moves after the pilot and G3 relies on internal security testing only.

1.11 Product risks

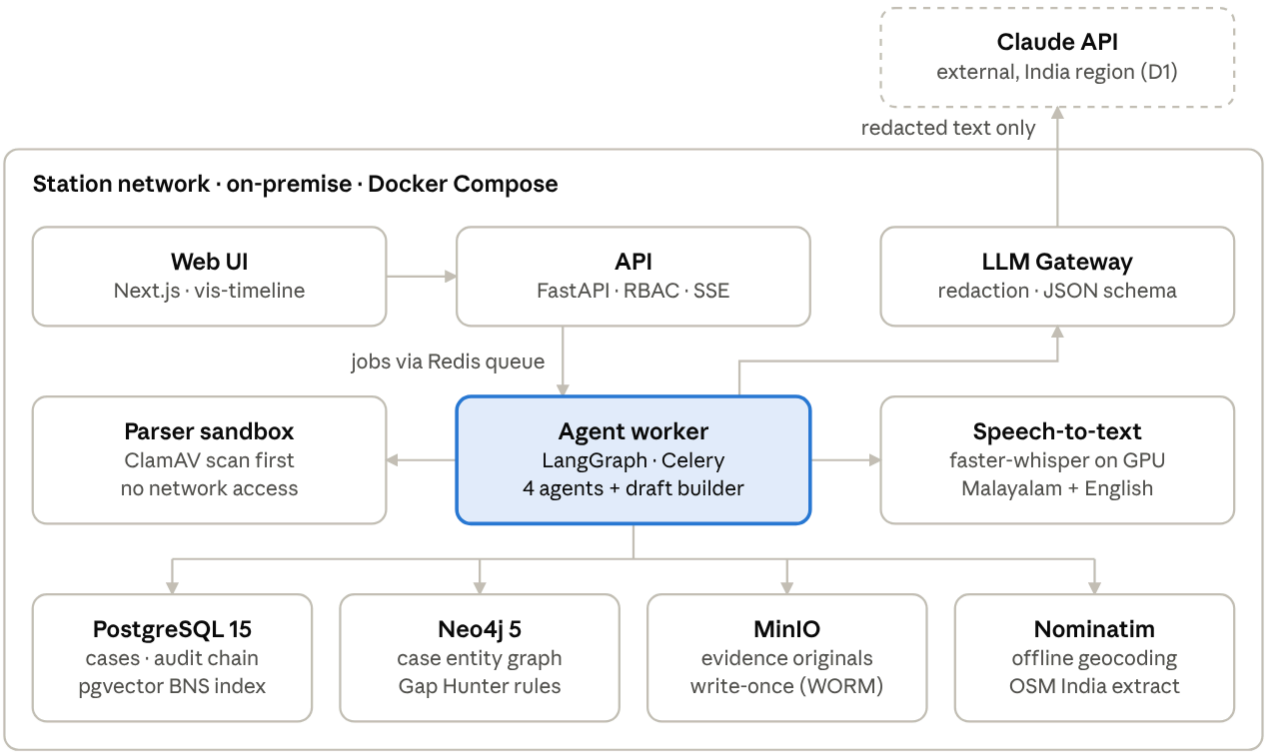
Risk	Likelihood	Impact	Mitigation
Officers see TRACE as extra work	Medium	High	Co-design the interface with station staff in Months 3–4; manual fallback at every step; measure time saved from day one
False manipulation flags embarrass genuine complainants	Medium	High	Flags go only to the officer and lab, never to the complainant; neutral wording; publish the false-positive rate

Risk	Likelihood	Impact	Mitigation
Poor Malayalam transcription	High	High	Benchmark in Month 1 (D7); officer correction in the interface; keep the original audio
Data-residency objection to the cloud LLM	High	Blocking	Decision D1: LLM gateway, redacted text only, on-premise fallback
Wrong BNS suggestion relied on	Medium	High	Advisory only, with the statute text and matching facts shown; the SHO must accept each one
Pilot access slips	Medium	Medium	Formalise the Cyberdome relationship into a written pilot MoU by Month 4

Part 2 — Technical Requirements Document (TRD)

2.1 Architecture overview

TRACE is a single-agency system packaged with Docker Compose and run on-premise. A Next.js interface talks to a FastAPI backend. The backend hands analysis jobs to a worker that runs the LangGraph agent pipeline. Evidence files, the case database and the entity graph all stay inside the station network. The only traffic that leaves goes through the LLM Gateway, and that traffic is redacted text, depending on decision D1.



TRACE deployment · station services and the one outbound path

Only the LLM Gateway has a path out of the station network, and in local-model mode (the D1 fallback) even that path is closed. Parsers handle untrusted files in a sandbox that has no network access.

2.2 Technology stack

The table keeps the stack from the proposal. Rows marked **(added)** or **(changed)** are recommendations from this document.

Layer	Choice	Purpose	Notes
Frontend	Next.js 14, Tailwind, shadcn/ui, vis-timeline	Officer interface and chronology view	Malayalam fonts bundled locally; no external CDNs, so it works offline
API	FastAPI on Python 3.11, Pydantic v2	REST API, authentication, validation	Versioned under `/api/v1`

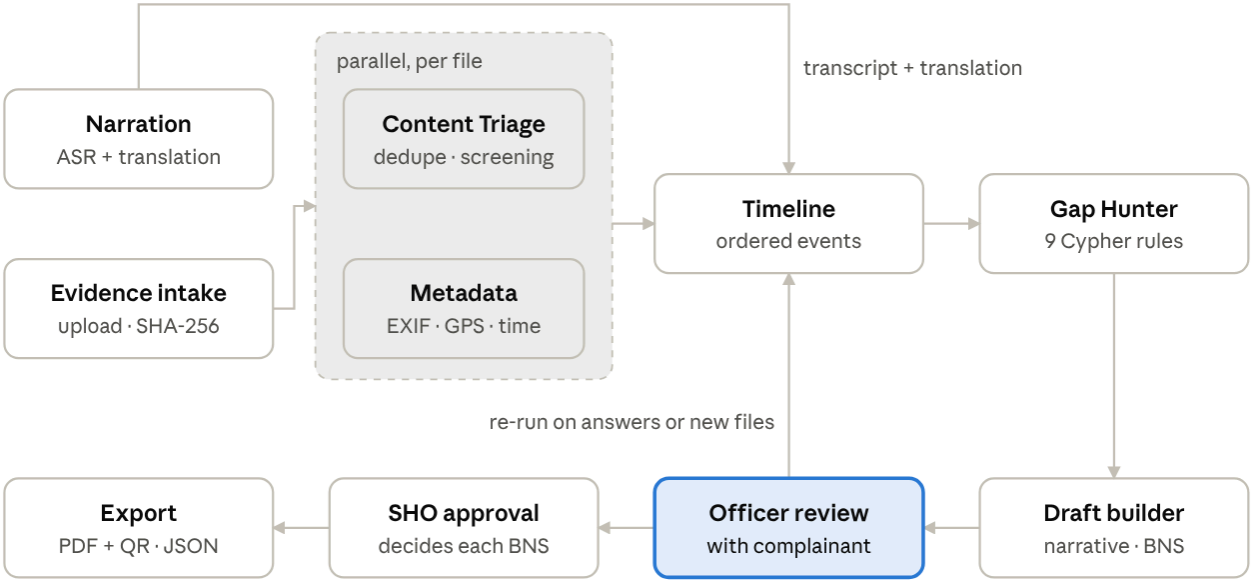
Layer	Choice	Purpose	Notes
Orchestration	LangGraph	Agent graph, state, retries, checkpoints	Checkpoints saved to Postgres, so a run can resume after a crash
Job queue **(added)**	Celery + Redis	Runs long jobs (speech-to-text, agents) outside API requests	Keeps API calls under 500 ms
LLM	Anthropic Claude, `claude-sonnet-5` by default, set in config (D2)	Narrative synthesis, translation, entity extraction, gap reasoning, BNS mapping	Called only through the LLM Gateway (§2.4)
Speech-to-text **(changed)**	Whisper large-v3 served with faster-whisper; Indic fine-tuned model as a candidate (D7)	Transcribe Malayalam and English narration and voice notes	faster-whisper is several times faster than reference Whisper on the same GPU
NER	spaCy (English) + LLM extraction	Entities from English text and translations	Malayalam text is translated first; entities are mapped back to the original by the LLM
Image forensics	OpenCV (DFT spectrum, noise residuals), plus ELA (error-level analysis) **(added)**	Manipulation screening	Calibrated on the D6 dataset; produces flags, never verdicts
Deduplication	imagehash (pHash, dHash)	Collapse near-duplicate screenshots	Default Hamming threshold 6, tuned on real bundles
Metadata	Pillow, piexif, plus pdfplumber **(added)** for bank-statement PDFs and mutagen **(added)** for audio tags	EXIF, GPS, timestamps, document text	

Layer	Choice	Purpose	Notes
Geocoding **(changed)**	**Self-hosted** Nominatim with an OpenStreetMap extract for India	Turn GPS coordinates into places	The public Nominatim server would send coordinates off-site and isn't allowed for bulk use
Language detection	langdetect, with fastText lid.176 as fallback	Route text and audio	langdetect is unreliable on short chat messages
Relational DB	PostgreSQL 15, SQLAlchemy 2, Alembic, pgvector **(added)**	Cases, audit chain, BNS retrieval index	
Graph DB	Neo4j 5 Community	Case entity graph for the Gap Hunter	One graph per case, separated by `case_id`
Evidence store **(added)**	MinIO with object lock (WORM: write once, read many)	Stores original files that can't be overwritten	Originals are never modified; derived files are kept separately
Malware scan **(added)**	ClamAV	Scan uploads before parsing	
PDF export	WeasyPrint + qrcode	Draft FIR and manifest PDFs with a QR code	
Observability **(added)**	Structured JSON logs, Prometheus, Grafana	Health, latency per agent, error rates	No personal data in logs

2.3 Agent pipeline

Triage and Metadata work file by file and don't depend on each other, so they run **in parallel**. Timeline needs both results. Gap Hunter runs on the finished timeline. When the officer adds answers or files, only the affected steps re-run.

Gaps are closed while the complainant is still at the counter



Agent pipeline · 4 agents, 1 officer loop

The officer's answers and any added files re-run only the affected steps. Filing happens outside TRACE, after export.

Shared state (LangGraph). Every agent reads and writes one typed `CaseState`: `case_id`, `run_id`, `evidence[]` (with type, hash, duplicate group, flags and metadata), `narrations[]` (original text, translation, segments), `events[]`, `entities[]`, `gaps[]`, `bns_candidates[]`, `errors[]`. Each agent returns a partial update. A checkpoint is saved after every node.

Agent	Input	Processing	Output	On failure or timeout
Content Triage	Original files	1) Check type by magic bytes; classify as financial record, conversation, media, call log, document or audio. 2) pHash and dHash, then group duplicates by Hamming distance. 3) For images: DFT spectrum peaks, JPEG double-compression signs, ELA and noise-residual consistency, combined into one score.	Evidence type, duplicate group with a canonical file, `screening_score` (0–1), `screening_reasons[]`, `referral_rank`	The file is marked "not screened" and the pipeline continues
Metadata	Original files	Read EXIF (DateTimeOriginal, Make/Model, Software, GPS), PDF metadata and audio tags. Reverse-geocode GPS offline. Compare against the claimed source: for example, a "phone camera photo" with Software=Photoshop, a timestamp after the upload time, or GPS far from the stated place.	`metadata{}`, `geo{}`, `conflicts[]` marked `absent` / `consistent` / `conflict`	The file is marked "metadata unavailable"

Agent	Input	Processing	Output	On failure or timeout
Timeline	Text from chats, SMS, statements and call logs; narration translation; metadata timestamps	Parse WhatsApp `.txt` exports and SMS formats with rules. The LLM pulls out events (who, what, when, amount, account, place) as JSON checked against a schema. Normalise dates to IST, merge duplicates, order events, and link each event to its source file and line.	`events[]` with `source_refs`, a confidence value and `time_precision` (exact, day or approximate); the entity graph is written to Neo4j	Rule-parsed events are kept and the officer sees a warning
Gap Hunter	Neo4j case graph, `events[]`	Run a fixed library of parameterised Cypher rules (§2.6). The LLM rewrites each result as a neutral question for the officer, with no judgement of credibility.	`gaps[]`: rule ID, severity, question in English (Malayalam optional), linked entities	Raw rule results are shown without the rewording
Draft builder (non-agent step)	Everything above	The LLM writes the chronological narrative using only `events[]`. Retrieval from a local BNS corpus (pgvector) returns candidate sections, and the LLM links each one to the facts that support it, quoting the section text.	Draft FIR JSON, BNS suggestions with citations, manifest, ranked referral list	Template draft from the events list

2.4 LLM Gateway

Every model call goes through one internal service. This is what lets decision D1 stay open without rewriting the agents.

- **Swappable provider:** `anthropic` (direct or cloud region) or `local` (an open-weight model run on-premise), chosen in configuration.
- **Redaction:** the gateway replaces phone numbers, account numbers, Aadhaar/PAN patterns and names with stable placeholders before sending, then restores them in the response. Raw images and audio are never sent.
- **Structured output:** every prompt declares a JSON schema, and responses that fail validation are retried once.
- **Timeouts:** 30 s per call with one retry, then the agent's fallback path (graceful degradation).
- **Evidence-text injection:** chat text from evidence is wrapped in delimiters marked as data. The system prompt tells the model never to follow instructions found in evidence. Outputs are checked against the schema, never executed.
- **Versioning:** prompts are stored as versioned templates. Every call logs the prompt version, model ID, token counts, latency, and SHA-256 hashes of the input and output (not the content itself) to the audit chain.
- **Cost control:** per-case token budget, plus prompt caching for the large fixed system prompts and the BNS corpus excerpts.

2.5 Data model

PostgreSQL (system of record)

Table	Key fields	Notes
`stations`	id, name, district, code	
`users`	id, station_id, name, badge_no, role, password_hash, status	Roles: writer, sho, io, analyst, admin
`cases`	id, station_id, ref_no, status, created_by, created_at, approved_by, approved_at	Status: draft → analysing → review → approved → exported
`complainants`	id, case_id, name, contact, language, consent_at, consent_text_version	Encrypted columns for personal data
`evidence_files`	id, case_id, filename, mime, size, sha256, object_key, uploaded_by, uploaded_at, evidence_type, dup_group_id, is_canonical	Originals stored in MinIO under object lock

Table	Key fields	Notes
`evidence_analysis`	file_id, run_id, screening_score, reasons jsonb, metadata jsonb, geo jsonb, conflicts jsonb	One row per run
`narrations`	id, case_id, audio_file_id, lang, transcript_orig, transcript_en, segments jsonb, corrected_by	Original and English both kept
`events`	id, case_id, run_id, occurred_at, precision, actor, action, amount, currency, place, source_refs jsonb, confidence	
`gaps`	id, case_id, rule_id, severity, question, status (open / answered / dismissed), answer, dismissed_reason	A reason is required to dismiss a gap
`bns_suggestions`	id, case_id, section, title, supporting_event_ids, statute_excerpt, decision, decided_by	Decision: accepted / rejected / pending
`drafts`	id, case_id, version, narrative_en, narrative_ml, body jsonb, created_by	Every save is a new version
`referrals`	id, case_id, file_id, rank, reason, confidence	
`audit_log`	seq, case_id, actor_id, action, payload_hash, prev_hash, entry_hash, at	Append-only; the database role has INSERT only
`pipeline_runs`	id, case_id, trigger, status, started_at, finished_at, node_status jsonb	

Neo4j (one graph per case, analysis only, rebuilt from Postgres)

- Nodes: `Person`, `Phone`, `Account`, `Device`, `Location`, `Event`, `Transaction`, `EvidenceFile`, `Statement` (a narration segment).
- Relationships: `(:Person)-[:OWNS]->(:Phone|:Account|:Device)`, `(:Event)-[:INVOLVES]->(:Person)`, `(:Transaction)-[:FROM]->(:Account)`, `(:Transaction)-[:TO]->(:Account)`, `(:Event)-[:AT]->(:Location)`, `(:Event)-[:EVIDENCED_BY]->`

`(:EvidenceFile)`, `(:Statement)-[:MENTIONS]->(*)`, `(:Event)-[:NEXT]->(:Event)`.

- Every node carries `case_id`. Every query filters on it.

2.6 Gap Hunter rule library

The rules are fixed, reviewed Cypher queries that take parameters. **The LLM never writes Cypher that gets executed.** This is the main defence against Cypher injection.

Rule	Detects	Severity
G-01	A transaction mentioned in the narration with no supporting financial record	High
G-02	A named person with no phone, account or address	Medium
G-03	Delay between the last incident event and the report date over a threshold (default 7 days), with no reason recorded	Medium (shown as a question, never an accusation)
G-04	One device or phone at two locations closer in time than travel would allow (speed over 900 km/h)	High
G-05	Money moved to an account with no bank or IFSC recorded	High
G-06	An event with only an approximate time where a source with an exact time may exist (for example, a bank SMS)	Low
G-07	A suspect contact (phone, UPI ID or URL) that appears in evidence but not in the narration	Medium
G-08	A file referenced in the narration ("he sent a photo") with no matching evidence	Medium
G-09	Amounts that don't add up: the stated total loss differs from the sum of transactions	High

Example (G-01):

```
MATCH (s:Statement {case_id: $case_id})-[:MENTIONS]->(t:Transaction)
WHERE NOT (t)-[:EVIDENCED_BY]->(:EvidenceFile)
RETURN t.id AS txn, t.amount AS amount, t.occurred_at AS at
```

2.7 Integrity: hash chain and verification

Each case has its own append-only chain of hashes. Every upload, analysis run, edit, decision, approval and export becomes one entry.

```
h_0 = \mathrm{SHA256}(\texttt{"TRACE"} \parallel station\_id \parallel
case\_id), \qquad h_i = \mathrm{SHA256}\big(h_{i-1} \parallel \mathrm{canon}
(e_i)\big)
```

- `canon(e)` means the entry encoded as canonical JSON (RFC 8785): sorted keys and UTF-8, so the same entry always produces the same hash.
- An evidence entry includes the SHA-256 of the original file, computed while the upload streams in, before anything else reads the file.
- **Daily station digest:** a scheduled job hashes the latest hash (head) of every active case into a station digest. The SHO signs it in the interface with the station key (Ed25519) and it is printed into the station register. This guards against an administrator rewriting a whole chain.
- **QR on export:** encodes `case_ref`, `export_version`, the head hash and an internal verification URL. Verification recomputes the chain from Postgres and the hashes of the MinIO originals.
- This is deliberately not a blockchain: one agency, one trust domain, simple to operate.

2.8 API design

REST over HTTPS, JSON, under `/api/v1`. Authentication uses short-lived access tokens with refresh tokens in httpOnly cookies. Long jobs return `202 Accepted` with a `run_id`; the interface follows progress through Server-Sent Events (SSE).

Method	Path	Purpose	Roles
POST	`/auth/login`, `/auth/refresh`, `/auth/logout`	Sessions	All
POST / GET	`/cases`, `/cases/{id}`	Create a case, read one	writer, sho
POST	`/cases/{id}/consent`	Record complainant consent	writer

Method	Path	Purpose	Roles
POST	`/cases/{id}/evidence`	Multipart upload, one or more files	writer
GET	`/cases/{id}/evidence`	Manifest with analysis results	writer, sho, io, analyst
GET	`/evidence/{fid}/content`	Stream the original (logged in the audit chain)	writer, sho, io, analyst
POST	`/cases/{id}/narrations`	Upload or record narration audio	writer
PATCH	`/narrations/{nid}`	Officer correction of transcript or translation	writer
POST	`/cases/{id}/runs`	Start or re-run the pipeline (`scope` : all or partial)	writer
GET	`/runs/{rid}/events`	SSE progress stream	writer, sho
GET	`/cases/{id}/timeline`	Ordered events with sources	all case roles
GET / PATCH	`/cases/{id}/gaps` , `/gaps/{gid}`	List gaps; answer or dismiss with a reason	writer, sho
GET / PUT	`/cases/{id}/draft`	Read the draft; save a new version	writer, sho
POST	`/bns-suggestions/{sid}/decision`	Accept or reject a suggestion	sho
POST	`/cases/{id}/approve`	SHO approval (locks the draft version)	sho
POST	`/cases/{id}/exports`	Produce PDF and JSON exports, manifest, referral sheet	sho
GET	`/cases/{id}/audit` , `/cases/{id}/audit/verify`	Audit trail and chain verification	sho, admin
GET	`/verify/{case_ref}/{head_hash}`	QR verification page	Network-internal

Method	Path	Purpose	Roles
CRUD	`/admin/users`	User management	admin

Errors follow RFC 9457 (`application/problem+json`). Every write takes an `Idempotency-Key` header.

2.9 Security

- **Authentication:** local accounts with argon2id password hashes, 8+ characters plus a lockout after 5 failed attempts. LDAP/AD integration is optional if the police network provides it. Idle sessions time out after 15 minutes.
- **Authorisation:** role-based access checked on every endpoint, plus station scoping: a user only sees their own station's cases. Analysts see only referred files.
- **Encryption:** TLS 1.2 or later inside the deployment (internal certificate authority). LUKS full-disk encryption on the host. Personal-data columns are encrypted at the application level (AES-256-GCM, keys in a local secrets file mounted read-only, upgradeable to a hardware security module later).
- **Upload handling:** extension and MIME allowlist, magic-byte check, 50 MB per file, 500 MB per case, a ClamAV scan, and decompression limits for ZIP chat exports (no nested archives, at most 1,000 entries, expansion ratio under 100:1). Parsers run in a separate worker container with no network access.
- **Graph:** parameterised Cypher only. The graph database user can only run the rule procedures.
- **LLM:** redaction, a filter on evidence text, schema-checked outputs, and no tool or code execution (§2.4).
- **Logging:** no personal data or evidence content in application logs. The audit chain records hashes and IDs.
- **Supply chain:** pinned dependencies, SBOM (software bill of materials) generated in CI, container image scanning, and offline update bundles signed by the vendor.
- **Assessment:** OWASP ASVS Level 2 as the target. The external audit from the budget line happens before multi-station rollout.

2.10 Testing and evaluation

Layer	What	Pass bar (proposed)
Unit	Parsers, hashing, Cypher rules, redaction	80% or more line coverage on core modules

Layer	What	Pass bar (proposed)
Integration	API to worker to databases, run inside Docker Compose in CI	All critical paths green
Agent golden set	25 anonymised synthetic cases, each with a known timeline, gaps and sections	Timeline event recall 90% or more; gap-rule recall 85% or more; zero invented events in the narrative
Manipulation calibration	D6 dataset, including WhatsApp-recompressed images	False-positive and false-negative rates published per class; nothing claimed beyond what was measured
Speech-to-text	30+ consented complaint recordings	WER 25% or less (Malayalam); named-entity accuracy reported
Security	SAST, dependency scan, ZAP baseline, manual tests for prompt injection and Cypher injection	No high or critical findings open at pilot
Usability	Five-officer test sessions in Months 3 and 4	SUS 70 or more

2.11 Deployment

- **Packaging:** one `docker-compose.yml` with these services: `web`, `api`, `worker` (agents), `asr` (GPU), `parser` (sandboxed), `postgres`, `neo4j`, `redis`, `minio`, `nominatim`, `clamav`, `llm-gateway`, `prometheus`, `grafana`, and a reverse proxy (Caddy or nginx).
- **Reference station hardware (proposed, to be checked by load test):** one server with 16 CPU cores, 64 GB RAM, one 16–24 GB NVIDIA GPU, 2 TB of mirrored SSD, and a UPS. Without a GPU, transcription runs on CPU and is about 5–10 times slower.
- **Network:** reachable only on the station LAN or police intranet. The only outbound connection is from `llm-gateway` (if D1 chooses cloud), through an allowlisted proxy.
- **Backups:** nightly encrypted Postgres dump, MinIO replication to a second disk, a monthly restore drill; RPO 24 h and RTO 8 h (how much data may be lost, and how long recovery may take).
- **Updates:** signed offline bundles, applied by the vendor with the station administrator present; Alembic migrations, with rollback tested.

2.12 Performance budgets

Operation	Budget (reference hardware)
API read (p95)	300 ms or less
Upload acknowledgement with hash, 50 MB file	5 s or less
Transcription, 10 minutes of audio (GPU)	2 min or less
Full pipeline, 50 files plus 10 minutes of narration	6 min or less, with first results streamed within 60 s
Gap re-run after the officer answers	30 s or less
PDF export	10 s or less

Part 3 — Software Requirements Specification (SRS)

3.1 Introduction

Purpose. This SRS lists the testable requirements for TRACE v1. Each requirement has an ID, a priority and an acceptance criterion. QA test cases and the security audit refer to these IDs.

Scope. TRACE v1 handles intake, analysis, drafting, review, approval and export of a draft FIR at one police station. It does not file FIRs, connect to CCTNS, or support investigation after registration (PRD §1.3).

References. PRD (Part 1), TRD (Part 2), BNS 2023, BNSS 2023, BSA 2023 (Appendix B), OWASP ASVS 4.0, RFC 8785, RFC 9457.

3.2 Overall description

- **Product perspective:** a standalone, on-premise web application. Its output is handed to the official FIR channel by a person.
- **User classes:** FIR writer, SHO, IO, forensic analyst, administrator (PRD §1.4).
- **Operating environment:** Chromium-based browsers or Firefox (last 2 versions) on station desktops at 1366×768 or larger. The server is a Linux host running Docker (TRD §2.11).
- **Constraints:** on-premise only; data stays in India; no auto-filing; English output, with Malayalam input and the Malayalam original kept; procurement through GeM, which means audit and documentation must be complete.

- **Assumptions:** the station has a LAN, a UPS, and a microphone or headset at the counter. The complainant gives consent to processing. The agency approves the LLM hosting option (D1).
- **Dependencies:** LLM provider (D1/D2), the speech model (D7), the OSM India extract, and a BNS statute text corpus in machine-readable form.

3.3 Functional requirements

FR-AUTH — Accounts and access

ID	Requirement	Pri.	Acceptance criterion
FR-AUTH-01	The system shall authenticate users with a username and password.	M	A valid login returns a session. An invalid one returns a generic error without saying which field was wrong.
FR-AUTH-02	The system shall lock an account for 15 minutes after 5 consecutive failed logins.	M	The 6th attempt within the lockout window is rejected and logged.
FR-AUTH-03	The system shall enforce role-based permissions per TRD §2.8.	M	Every endpoint returns 403 for a role it doesn't allow (automated matrix test).
FR-AUTH-04	The system shall restrict each user to the cases of their own station.	M	A cross-station case ID returns 404.
FR-AUTH-05	The system shall end sessions after 15 minutes of inactivity.	M	A request after 15 minutes idle needs a new login.

FR-CASE — Case lifecycle

ID	Requirement	Pri.	Acceptance criterion
FR-CASE-01	The system shall create a case with a unique station-scoped reference number.	M	Reference format `<STATION>-<YYYY>-<NNNNN>`, unique under concurrent creation.
FR-CASE-02	The system shall record the complainant's identity, contact, preferred language and consent before evidence can be uploaded.	M	Upload is blocked until consent is recorded; consent text version is saved.

ID	Requirement	Pri.	Acceptance criterion
FR-CASE-03	The system shall move cases through the states draft → analysing → review → approved → exported, allowing only defined transitions.	M	An illegal transition returns 409.
FR-CASE-04	The system shall make an approved case read-only, except for a new export or an SHO-initiated reopen with a stated reason.	M	Edits after approval return 409; a reopen is logged with its reason.

FR-INT — Evidence intake

ID	Requirement	Pri.	Acceptance criterion
FR-INT-01	The system shall accept multi-file uploads of these types: JPG, PNG, WEBP, HEIC, PDF, TXT, CSV, XLSX, ZIP (chat export), OGG/OPUS, MP3, M4A, WAV.	M	Each allowed type uploads. A disallowed type is rejected with a reason.
FR-INT-02	The system shall reject files over 50 MB and cases over 500 MB in total.	M	A 50.1 MB file is rejected; the officer sees the limit.
FR-INT-03	The system shall check each file's type by its content (magic bytes) and reject a file whose content doesn't match its extension.	M	A renamed `.exe` saved as `.jpg` is rejected.
FR-INT-04	The system shall scan every file for malware before parsing it.	M	The EICAR test file is quarantined and logged.
FR-INT-05	The system shall compute the SHA-256 of each original file while it uploads and write the hash to the case's audit chain before any other processing.	M	The stored hash equals `sha256sum` of the source file, and the audit entry precedes any analysis entry.
FR-INT-06	The system shall store originals so they can't be overwritten, and keep all derived files apart from them.	M	An attempt to overwrite an original object fails.

ID	Requirement	Pri.	Acceptance criterion
FR-INT-07	The system shall expand WhatsApp ZIP exports within set limits (no nested archives, at most 1,000 entries, expansion ratio under 100:1).	M	A zip-bomb test file is rejected.
FR-INT-08	The system shall let the officer add files at any point before approval.	M	Files added later trigger a partial re-run.

FR-NAR — Narration and translation

ID	Requirement	Pri.	Acceptance criterion
FR-NAR-01	The system shall record narration in the browser or accept it as an uploaded audio file.	M	A 10-minute recording saves and is hashed.
FR-NAR-02	The system shall detect whether the narration is Malayalam or English and transcribe it.	M	Language is detected correctly on 95% or more of the benchmark set.
FR-NAR-03	The system shall translate a Malayalam transcript into English and show the original and the translation side by side, aligned segment by segment.	M	Every English segment links to its Malayalam segment and audio time range.
FR-NAR-04	The system shall let the officer correct the transcript and translation, keeping the machine version and the edit history.	M	Both versions can be retrieved; the edit is in the audit chain.
FR-NAR-05	The system shall keep names, numbers and amounts as they appear in the original, transliterated, rather than translated.	M	Golden-set check: names and amounts are unchanged in 98% or more of cases.
FR-NAR-06	The system shall transcribe voice notes found in the evidence the same way.	S	Voice notes appear as conversation evidence with a transcript.

FR-TRI — Content Triage Agent

ID	Requirement	Pri.	Acceptance criterion
FR-TRI-01	The system shall classify each file as one of: financial record, conversation, media, call log, document, audio, other.	M	Classification accuracy of 90% or more on the golden set.
FR-TRI-02	The system shall group near-duplicate images by perceptual hash and mark one canonical file per group.	M	95% or more of the known duplicates in the golden set are grouped, with under 2% false grouping.
FR-TRI-03	The system shall keep every duplicate file and its hash. Grouping changes only how files are displayed.	M	All originals stay listed in the full manifest.
FR-TRI-04	The system shall compute a manipulation-screening score (0–1) and reasons for each image.	M	Score and reasons are present for every image or the file is marked "not screened".
FR-TRI-05	The system shall present screening results only as "recommend forensic review" or "no screening concern", with confidence and a fixed statement of the method's limits.	M	The UI never uses the words "fake", "forged" or "authentic"; the limitations text appears on every flag.
FR-TRI-06	The system shall produce a ranked forensic referral list from the screening score, the evidence's importance to the timeline, and any metadata conflicts.	M	The list is ordered and each item carries a reason.
FR-TRI-07	The system shall screen audio for signs of synthetic speech.	C	— (post-pilot)

FR-MET — Metadata Agent

ID	Requirement	Pri.	Acceptance criterion
FR-MET-01	The system shall extract EXIF (capture time, device, software, GPS), PDF metadata and audio tags where present.	M	Fields match `exiftool` output on the test set.

ID	Requirement	Pri.	Acceptance criterion
FR-MET-02	The system shall reverse-geocode GPS coordinates using the self-hosted geocoder only.	M	No outbound geocoding traffic during tests (checked against the network egress log).
FR-MET-03	The system shall mark each metadata check as absent, consistent or conflict, and show absence neutrally.	M	A WhatsApp-stripped image shows "metadata not available" and isn't flagged.
FR-MET-04	The system shall flag a conflict when the capture time is later than the upload time, the editing-software tag doesn't fit the claimed source, or the GPS location contradicts the stated place by more than a set distance.	M	Each conflict type triggers on its test fixture.

FR-TML — Timeline Agent

ID	Requirement	Pri.	Acceptance criterion
FR-TML-01	The system shall extract events (time, actor, action, amount, account, place) from chats, SMS, statements, call logs, narration and metadata.	M	Event recall of 90% or more on the golden set.
FR-TML-02	The system shall link every event to at least one source (file and line or position, or narration segment).	M	100% of events have a `source_ref`; clicking it opens the source.
FR-TML-03	The system shall normalise all times to IST and record their precision (exact, day or approximate).	M	Mixed-format fixtures normalise correctly.
FR-TML-04	The system shall merge events that describe the same occurrence from several sources, and show all the sources.	M	A bank SMS and a statement line for the same transfer become one event with two sources.
FR-TML-05	The system shall display the timeline in vis-timeline with filters by type and person, and zoom.	M	200 events render in under 1 s.

ID	Requirement	Pri.	Acceptance criterion
FR-TML-06	The system shall not create any event that lacks a supporting source.	M	Zero events without a source on the golden set.

FR-GAP — Gap Hunter Agent

ID	Requirement	Pri.	Acceptance criterion
FR-GAP-01	The system shall run rules G-01 to G-09 (TRD §2.6) after every timeline build.	M	Each rule fires on its fixture and stays silent on the clean fixture.
FR-GAP-02	The system shall run only reviewed, parameterised Cypher queries, never Cypher written by the model.	M	Code review and test: injected Cypher in evidence text has no effect.
FR-GAP-03	The system shall phrase each gap as a neutral question addressed to the officer.	M	A wording linter blocks a list of judgement words ("lying", "suspicious", "inconsistent story", "credibility" and similar).
FR-GAP-04	The system shall never compute or display any credibility, truthfulness or deception score.	M	No such field exists in the schema or UI (design review).
FR-GAP-05	The system shall let the officer answer a gap (by text, voice or new file) or dismiss it with a required reason.	M	A dismissal without a reason is rejected.
FR-GAP-06	The system shall re-run the affected agents after an answer and update gap statuses.	M	The re-run finishes within 30 s on reference hardware.
FR-GAP-07	The system shall show unresolved gaps on the SHO's approval screen.	M	Approval with open high-severity gaps needs an explicit acknowledgement.

FR-DRF — Draft FIR and BNS suggestions

ID	Requirement	Pri.	Acceptance criterion
FR-DRF-01	The system shall generate an English narrative in chronological order using only extracted events.	M	Every sentence maps to at least one event ID (hidden trace data); no invented facts on the golden set.
FR-DRF-02	The system shall include an evidence manifest: file, type, hash, duplicate group, screening result and metadata status.	M	Manifest rows equal the number of uploaded files.
FR-DRF-03	The system shall suggest BNS sections, each with the statute text, the supporting facts, and the label "Advisory: for officer decision".	M	Every suggestion has an excerpt and at least one event link.
FR-DRF-04	The system shall suggest only sections found in the local BNS corpus.	M	Any section outside the corpus is discarded and logged.
FR-DRF-05	The system shall require the SHO to accept or reject each suggestion before approval.	M	Approval is blocked while any suggestion is pending.
FR-DRF-06	The system shall let officers edit the draft freely, saving each save as a new version.	M	Version history can be shown with a diff.
FR-DRF-07	The system shall never submit, file or transmit the FIR to any outside system.	M	No submission endpoint exists; egress test passes.
FR-DRF-08	The system shall produce a Malayalam rendering of the final narrative for reading back to the complainant.	S	Rendering is available after approval.

FR-AUD — Chain of custody and audit

ID	Requirement	Pri.	Acceptance criterion
FR-AUD-01	The system shall append an audit entry for every upload, view of an original, analysis run, edit, decision, approval, export and login.	M	A scripted session produces the expected entries.
FR-AUD-02	The system shall chain audit entries with SHA-256 as defined in TRD §2.7.	M	Recomputing the chain matches the stored hashes.
FR-AUD-03	The system shall detect and report any altered, deleted or reordered entry.	M	Tampering with one row in the test database makes verification fail at that row.
FR-AUD-04	The system shall produce a signed daily station digest of all case head hashes.	S	The digest signature verifies with the station's public key.
FR-AUD-05	The database shall allow only appends to the audit table.	M	UPDATE and DELETE are denied for the application role.

FR-EXP — Export and verification

ID	Requirement	Pri.	Acceptance criterion
FR-EXP-01	The system shall export the approved draft as PDF (A4, English plus Malayalam annex) and JSON.	M	Files open in standard readers; JSON validates against its schema.
FR-EXP-02	The system shall print a QR code on the PDF that encodes the case reference, export version and head hash.	M	Scanning it on the station network opens the verification result.
FR-EXP-03	The system shall export the forensic referral sheet with file hashes, ranks and reasons.	M	Hashes on the sheet match the stored originals.
FR-EXP-04	The system shall pre-fill hash and device details for an electronic-evidence certificate under BSA Section 63 (verify).	S	Template fields are populated from the manifest.

FR-ADM — Administration

ID	Requirement	Pri.	Acceptance criterion
FR-ADM-01	The administrator shall create, edit, disable and re-enable users and assign roles.	M	A disabled user can't log in; history is kept.
FR-ADM-02	The system shall show service health and pipeline error rates to administrators.	S	The dashboard shows status for every service.
FR-ADM-03	The system shall let administrators set retention periods and run purges, with the case number and an approval recorded.	S	A purge leaves a tombstone entry in the audit chain.

3.4 Non-functional requirements

ID	Category	Requirement	Pri.
NFR-PERF-01	Performance	Meet the budgets in TRD §2.12 on reference hardware.	M
NFR-PERF-02	Performance	Support 5 concurrent officers per station without breaching the budgets.	M
NFR-REL-01	Reliability	99.5% availability during station working hours, measured monthly.	M
NFR-REL-02	Reliability	A failed or timed-out model call shall never block manual completion of the FIR.	M
NFR-REL-03	Reliability	Resume a pipeline from its last checkpoint after a worker restart.	M
NFR-REL-04	Reliability	RPO 24 h and RTO 8 h, with a monthly restore drill.	M
NFR-SEC-01	Security	Meet OWASP ASVS Level 2.	M
NFR-SEC-02	Security	TLS 1.2 or later on all internal traffic; encrypted disks; personal-data columns encrypted with AES-256-GCM.	M
NFR-SEC-03	Security	The only outbound network path is the LLM Gateway through an allowlisted proxy (or none, in local mode).	M

ID	Category	Requirement	Pri.
NFR-SEC-04	Security	No personal data or evidence content in application logs.	M
NFR-SEC-05	Security	Prompt-injection and Cypher-injection test suites pass before each release.	M
NFR-PRIV-01	Privacy	Only redacted text is sent to an external LLM; raw media is never sent.	M
NFR-PRIV-02	Privacy	All data is stored in India, inside the agency's environment.	M
NFR-PRIV-03	Privacy	Consent is recorded with the version of the notice the complainant was shown.	M
NFR-USE-01	Usability	A new officer completes a standard case after 1 hour of training.	M
NFR-USE-02	Usability	SUS score of 70 or more in pilot surveys.	S
NFR-USE-03	Usability	WCAG 2.1 AA for contrast and keyboard navigation.	S
NFR-LOC-01	Localisation	The interface is in English with Malayalam labels available. Malayalam text renders correctly using locally bundled fonts.	M
NFR-MNT-01	Maintainability	Model IDs, prompt versions, thresholds and rule parameters are set in configuration, not in code.	M
NFR-MNT-02	Maintainability	80% or more unit-test coverage on core modules; CI runs on every merge.	M
NFR-AI-01	Responsible AI	Every AI output in the interface is labelled as AI-generated, with its source links.	M
NFR-AI-02	Responsible AI	Screening and gap results show stated limitations; no credibility scoring of any kind.	M
NFR-AI-03	Responsible AI	Before each release, publish the measured error rates (WER, event recall, screening false positives and false negatives) in an internal model card.	M

3.5 Use cases

ID	Use case	Actor	Main flow	Alternate flows
UC-01	Register a complaint with evidence	FIR writer	Open case → record consent → upload files → record narration → start analysis → watch progress	File rejected (type or size) → reason shown, others continue · transcription fails → officer types the narration
UC-02	Resolve gaps with the complainant present	FIR writer	Open the gap list → ask each question → record the answer or add a file → partial re-run → status updates	Complainant doesn't know → mark "answered: not known" · not relevant → dismiss with a reason
UC-03	Review and approve the draft	SHO	Review narrative → decide each BNS suggestion → review referrals → acknowledge open gaps → approve	Send back to the writer with a comment
UC-04	Export and file	SHO	Export the PDF and JSON → file through the official channel (outside TRACE) → record the official FIR number in TRACE	Re-export creates a new version; the earlier ones remain
UC-05	Verify an exported document	IO, auditor, court liaison	Scan the QR code → the system recomputes the chain → shows a match or a mismatch	Chain broken → mismatch at a specific entry, and an alert to the administrator
UC-06	Process a forensic referral	Forensic analyst	Open the referral queue → download originals (logged) → record the outcome per file	—

3.6 External interfaces

- **User interface:** responsive web app, desktop-first. Main screens: Case list, Intake, Narration, Analysis progress, Timeline, Gaps, Draft and BNS, Approval, Audit, Admin.
- **Hardware:** a USB microphone or headset; optionally a document scanner (images upload like any other file).
- **Software:** the LLM provider API through the gateway; the local Nominatim, ClamAV, MinIO, Neo4j and Postgres services.
- **Communications:** HTTPS only; SSE for progress updates; no inbound connections from outside the station network.

3.7 Traceability

PRD feature	Requirements	Verified by
F1 Intake	FR-CASE-01..04, FR-INT-01..08	Integration tests, security tests
F2 Narration	FR-NAR-01..06	WER benchmark, golden set
F3 Triage	FR-TRI-01..07	Golden set, D6 calibration report
F4 Metadata	FR-MET-01..04	Fixtures, egress test
F5 Timeline, F11 Timeline view	FR-TML-01..06	Golden set, UI performance test
F6 Gap Hunter	FR-GAP-01..07	Rule fixtures, wording linter, injection tests
F7 Draft FIR	FR-DRF-01..08	Golden set (no invented facts), approval tests
F8 Custody and audit	FR-AUD-01..05	Tamper tests
F9 Export	FR-EXP-01..04	Export and QR tests
F10 Users and roles	FR-AUTH-01..05, FR-ADM-01..03	Role-permission matrix test
F12 Certificate helper	FR-EXP-04	Legal review of the template

Appendices

Appendix A — Glossary

Term	Meaning
FIR	First Information Report: the first written record of a cognisable offence
e-FIR / Zero FIR	An FIR given by electronic means, or registered at any station regardless of where the offence happened
BNS / BNSS / BSA	Bharatiya Nyaya Sanhita, Bharatiya Nagarik Suraksha Sanhita and Bharatiya Sakshya Adhiniyam (all 2023), which replaced the IPC, CrPC and Evidence Act
SHO / IO	Station House Officer / Investigating Officer
CCTNS	Crime and Criminal Tracking Network and Systems, the national police records system
Cyberdome	Kerala Police's cyber security and research centre
Malkhana	The station's store for seized property and evidence
pHash / dHash	Perceptual hashes: fingerprints of an image that stay similar when it is resized or recompressed
DFT / ELA	Discrete Fourier Transform (frequency-domain analysis) / Error Level Analysis (JPEG recompression differences)
EXIF	Metadata embedded in an image file (capture time, device, GPS)
WER	Word error rate, the standard accuracy measure for speech-to-text
SUS	System Usability Scale, a 10-question usability score from 0 to 100
RPO / RTO	Recovery point objective (how much data may be lost) / recovery time objective (how long recovery may take)

Term	Meaning
GeM / KSUM / DPIIT	Government e-Marketplace / Kerala Startup Mission / Department for Promotion of Industry and Internal Trade

Appendix B — Legal references to verify

These references were written from memory and have **not been checked against the statute text**. A legal mentor should confirm each one before the proposal is submitted or the BNS corpus is built.

Reference	Believed to cover	Used in
BNSS 2023, Section 173	Information in cognisable cases, including information by electronic communication (e-FIR, to be signed within 3 days) and registration regardless of jurisdiction (Zero FIR)	Problem statement, D3
BNSS 2023, Section 176(3)	Forensic experts visiting crime scenes for offences punishable with 7 years or more	Forensic referral context
BSA 2023, Section 63	Admissibility of electronic records and the required certificate, which includes hash values	F12, FR-EXP-04
BNS 2023, Sections 318–319	Cheating; cheating by personation	BNS corpus, golden-set cases
IT Act 2000, Sections 66C and 66D	Identity theft; cheating by personation using a computer resource	BNS corpus: **cyber-fraud FIRs usually cite IT Act sections too, so the corpus should include them**
DPDP Act 2023	Personal-data processing and the exemptions for state agencies	Privacy design, consent notice

Appendix C — Other open questions

- [ ] Is the suggestion corpus BNS only, or BNS plus the IT Act and other special laws? (Recommendation: BNS plus the IT Act for v1.)
- [ ] Retention period for case data and originals after export, set by agency policy.
- [ ] Who signs the pilot MoU, and does it cover the data-processing terms for D1?
- [ ] Do the 30+ consented complaint recordings for the D7 benchmark need ethics or agency approval?
- [ ] Does the station have a GPU-capable server, or is it part of the ₹1.5L pilot hardware line?
- [ ] Should the legal-aid build (US-15) share the codebase with a feature flag, or be a separate product?